Akul Ramasubban

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Education

Purdue University ***West Lafayette, IN***
B.S. in Mechanical Engineering - Presidential Scholar ***Expected May 2027***

Technical Experience

Tesla
Cell Manufacturing Equipment/Process Design Engineering Intern ***January 2026 - August 2026***

- Designed a shuttle-carried conveyor cleaner with non-shedding foam for monthly preventive maintenance, reducing cell contamination, cutting maintenance downtime by **25%**, and shortening cleaning intervention time by **87.5%**.
- Qualified replacement fasteners for vision inspection tools using measured vibration and a custom rig simulating six months of service, reducing false rejects and improving tool availability by **~10%** after fleetwide rollout.
- Optimized laser-welding exhaust flow through root cause analysis and DOE in JMP to restore penetration and improve downstream yield by **~10,000** cells/week, validating the process with pull tests and cathode-dust checks.
- Standardized laser-weld power fleetwide and recalibrated each tool, reducing weld-strength asymmetry by **20%**.
- Built and deployed a four-tube current-collector alignment jig in one week using IMU-measured vibration and preset controls to correct stack misalignment, achieving zero pickup faults over **60,000** cells in production trials.
- Deployed a standardized compression-fit protection tray across winder generations after equipment interface surveys and three prototype iterations, reducing shuttle damage by **20%** and enabling same-day replacement.

Ford Motor Company
Product Development Intern - Forward Model ***May 2025 - August 2025***

- Implemented supplier pre-ship containment by coordinating with supplier teams after a Six Sigma 3L5Y root cause study of injector-tip cavitation that caused engine knock, preventing a potential field issue and an NHTSA recall.
- Established the team's engine-wide DFMEA in MagicDraw to organize failure modes and detection controls across thermal, NVH, and related system interfaces, providing a common framework for engine design-risk management.
- Expanded injector reliability test coverage through low-use/restart cycles and extended fatigue testing to reproduce cavitation-related failure conditions and define the specific reliability tests required for upcoming engine models.
- Mapped real-world customer usage to refine reliability test methods, contributing to **20%** higher in-field quality.

Purdue Electric Racing (Formula SAE EV)
Battery Senior Advisor ***August 2023 - January 2026***

- Redesigned the battery-module structure to increase specific energy by **30%** while redirecting inertial loads around cell cans and thermal pads, using hand calculations and rule-based checks to establish the primary load path.
- Engineered a thermal interface and spacer system to balance cooling, electrical isolation, and mechanical loading, improving battery performance by **10%** through pad compression control and worst-case tolerance analysis.
- Integrated cooling plates as structural pack walls to cut cooling-system weight by **11%** and pack width by **1 inch**, using insulating barriers to align cells and maintain electrical isolation during wire bonding and assembly.
- Tested three cooling-plate designs under simulated thermal loads and cell compression using a custom Arduino rig with eight thermistors and coolant-flow logging, measuring temperatures within **1.2** °C of CFD predictions.
- Created a waterproof battery enclosure that passed inspection without leaks, contributing to a top-10 team finish.

Purdue Humanoid Robot Club
Quadruped Project Lead ***August 2026 - Present***

- Direct mechanical design, actuator selection, and FEA for a quadruped with four actuated DOF and spring-assisted leg linkages, coordinating team design reviews and assembly to integrate actuation and onboard compute.
- Evaluate leg mechanisms on an 80/20 test stand and use ROS/Gazebo simulation and motor-firmware integration to support gait development in a prototype that walks unsupported, turns, and crouches with its own NUC.

Purdue MARS (Mechanisms and Robotic Systems) Lab
Undergraduate Researcher - NSF-funded project ***August 2026 - Present***

- Design a robot-arm-mounted apple gripper and cutter using a custom single-servo linkage with fixed and moving blades, combining compliant TPU grippers and a PETG structure to retain fruit throughout stem cutting.
- Automate grasping, laser exposure, and cutting in one sequence by collaborating with a software teammate to connect Unitree Go2 EDU perception outputs from SAM/DINO models to the cutter actuation interface.

Tredence Inc.
Data Science Intern ***May 2024 - August 2024***

- Developed multiple regression models for supply-chain data, contributing to **15%** lower costs for key customers.
- Analyzed retail data with Python, SQL, NumPy, and Matplotlib to identify trends and inform client decisions.

Skills

SolidWorks | 3DX | GD&T | FEA | Tolerance analysis | DFM | DOE | JMP | DFMEA | MATLAB | Python | SQL | ROS | Gazebo